```c
#include <stdio.h>

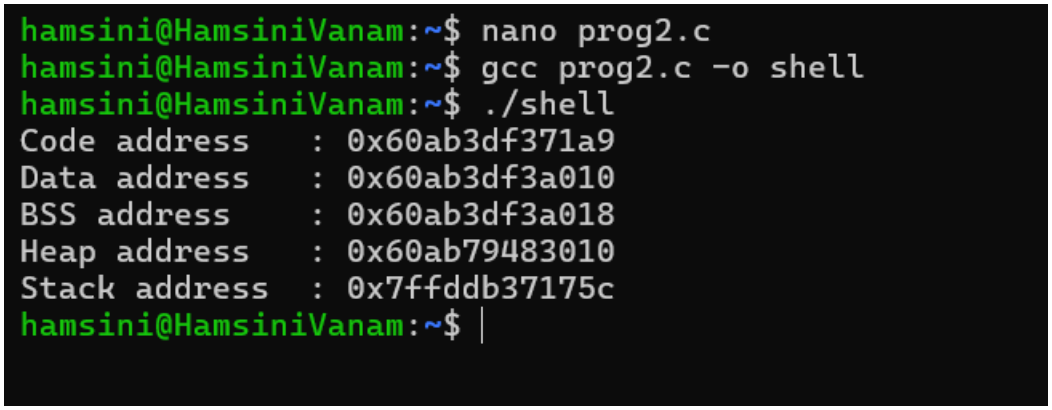

int main() {
   int x = 10;


   printf("Value of x = %d\n", x);
   printf("Virtual address of x = %p\n", (void *)&x);


   return 0;
}
```

```
hamsini@HamsiniVanam:~$ nano prog1.c
hamsini@HamsiniVanam:~$ gcc prog1.c -o shell
hamsini@HamsiniVanam:~$ ./shell
Value of x = 10
Virtual address of x = 0x7fffe5dfdae4
hamsini@HamsiniVanam:~$
```

```c
#include <stdio.h>
#include <stdlib.h>
int global = 10;
int bss_variable;
int main() {
int local = 20;
int *heap = malloc(sizeof(int));
*heap = 30;
printf("Code address : %p\n", (void *)main);
printf("Data address : %p\n", (void *)&global);
printf("BSS address : %p\n", (void *)&bss_variable);
printf("Heap address : %p\n", (void *)heap);
printf("Stack address : %p\n", (void *)&local);
free(heap);
return 0;
}
```




```
#include <stdio.h>

int main() {

   int page_table[4] = {1, -1, 3, -1};

   int page;

   printf("Enter page number (0-3): ");
   scanf("%d", &page);

   if (page_table[page] == -1) {
      printf("Page Fault!\n");
      printf("Page must be loaded into memory.\n");
   }
   else {
      printf("Page is already in memory.\n");
      printf("Physical frame = %d\n", page_table[page]);
   }

   return 0;
```

}

```
hamsini@HamsiniVanam:~$ nano prog3.c
hamsini@HamsiniVanam:~$ gcc prog3.c -o shell
hamsini@HamsiniVanam:~$ ./shell
Enter page number (0-3): 2
Page is already in memory.
Physical frame = 3
hamsini@HamsiniVanam:~$
```

```
#include <stdio.h>
int main() {

int page_table[] = {3, 5, 1, 7};

int page_size = 100;


int virtual_address = 250;


int page = virtual_address / page_size;

int offset = virtual_address % page_size;


int frame = page_table[page];


int physical_address = frame * page_size + offset;


printf("Virtual Address = %d\n", virtual_address);

printf("Page Number    = %d\n", page);

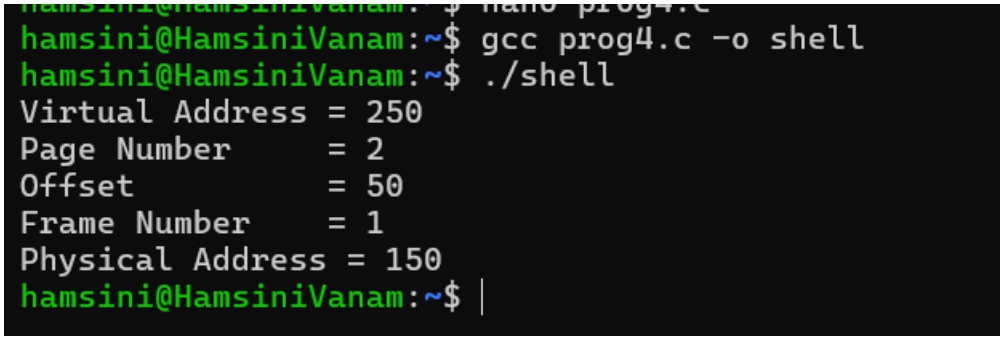
printf("Offset        = %d\n", offset);

printf("Frame Number    = %d\n", frame);

printf("Physical Address = %d\n", physical_address);


return 0;


}
```




```
int *p;


p = malloc(5 * sizeof(int));


if (p == NULL) {
   printf("Memory allocation failed\n");
   return 1;
}


for (int i = 0; i < 5; i++) {
   p[i] = (i + 1) * 10;
}


for (int i = 0; i < 5; i++) {
   printf("%d ", p[i]);
}


printf("\n");


free(p);


return 0;
```

```
hamsini@HamsiniVanam:~$ nano prog5.c
hamsini@HamsiniVanam:~$
hamsini@HamsiniVanam:~$ gcc prog5.c -o shell
hamsini@HamsiniVanam:~$ ./shell
10 20 30 40 50
hamsini@HamsiniVanam:~$
```

```
#include <stdio.h>

int main() {

int *p = NULL;

printf("Before accessing memory\n");

*p = 100;

printf("After accessing memory\n");

return 0;
}
```

```
hamsini@HamsiniVanam:~$ nano prog6.c
hamsini@HamsiniVanam:~$ gcc prog6.c -o shell
hamsini@HamsiniVanam:~$ ./shell
Before accessing memory
Segmentation fault         ./shell
hamsini@HamsiniVanam:~$
```